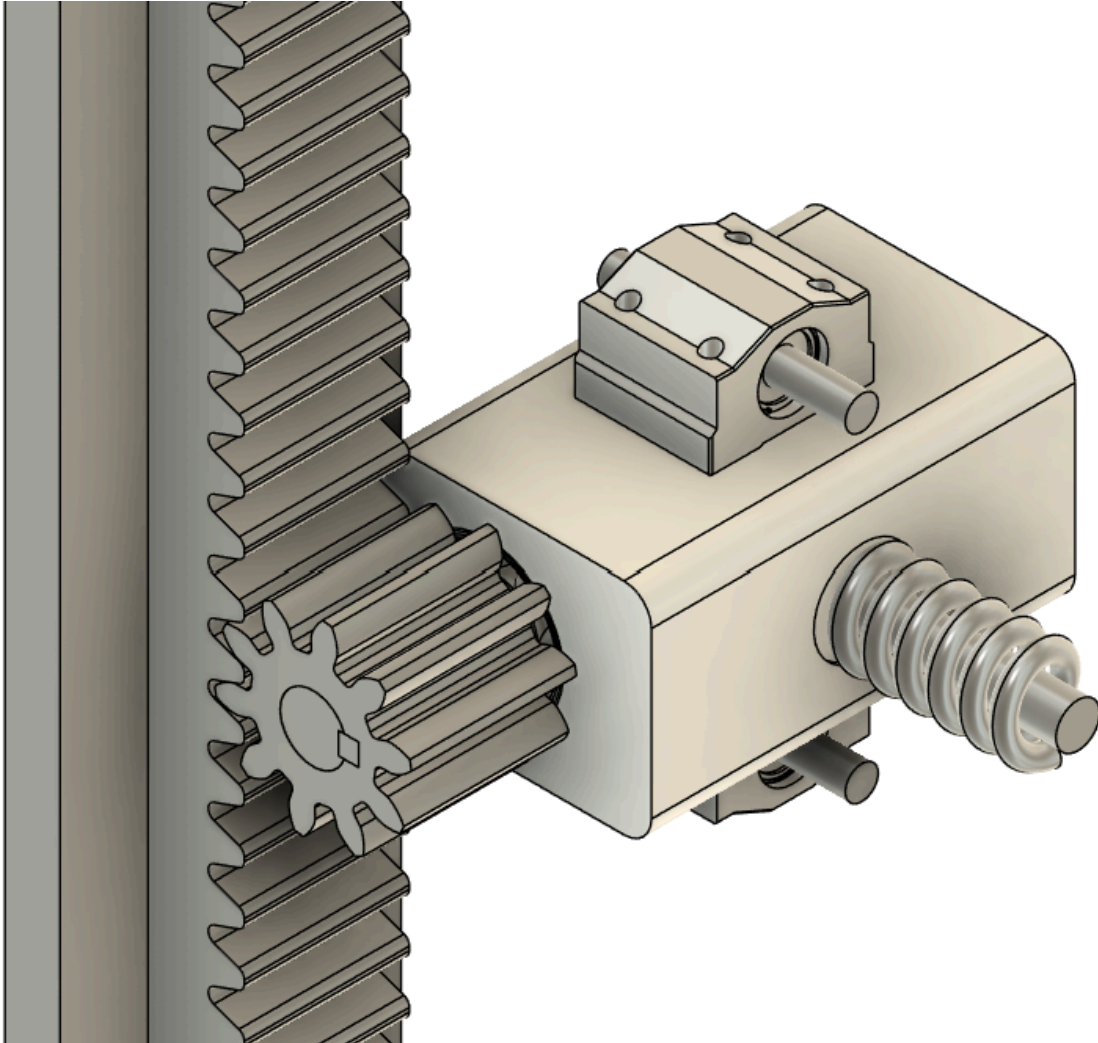


Vertical Climbing Robot System for Automated Warehousing



By Sreeram Padmesh

Pinion Aligner Mechanism

The Pinion Aligner Mechanism is a critical, self-adjusting subsystem designed to ensure precise and continuous meshing of the pinion and rack, compensating for manufacturing and installation tolerances across the 8-foot vertical track. This is achieved using a **spring-loaded axle system**.

The Sliding Block Assembly

The mechanism relies on a single **sliding block** to house and constrain the drive assembly:

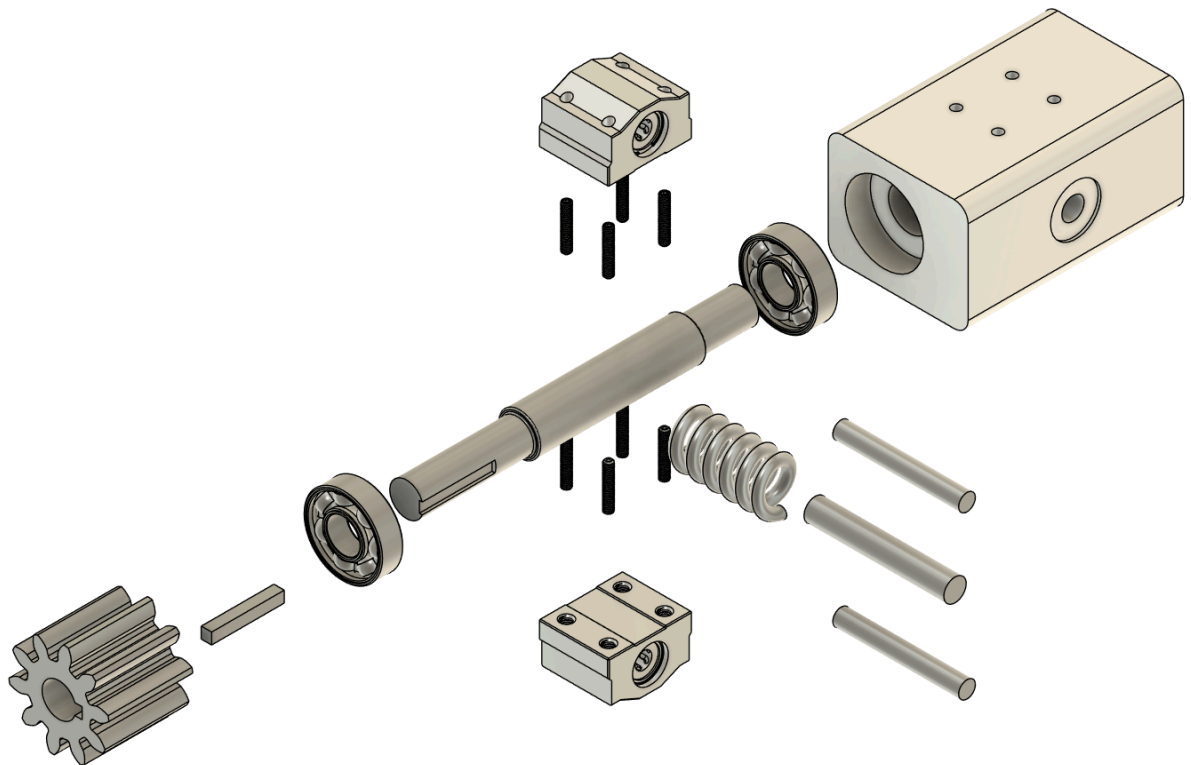
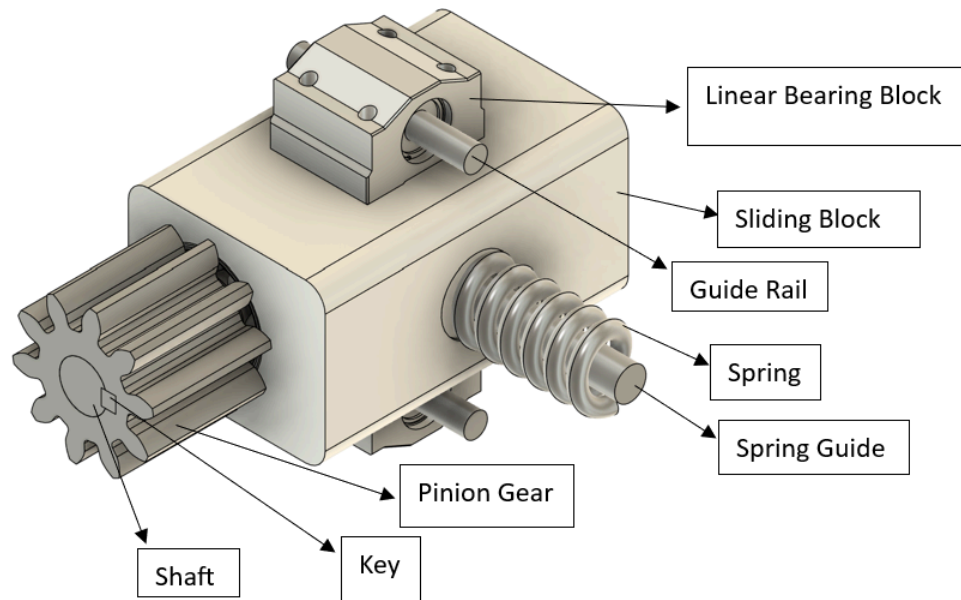
- **Axle Seating:** The 12 mm axle is supported by two 6001 series bearings (**12 mm ID, 28 mm OD and 8 mm in length**) which are press-fit into precision bores within the block. The bearings are separated by a **60 mm bearing span** to maximise resistance to bending moments induced by the lift load.
- **Translational Constraint:** The sliding block is mounted to the robot's main chassis via two parallel **linear guide rails**. This interface restricts the block to a single degree of freedom, translation **perpendicular** to the rack face, preventing any undesirable wobble or rotational movement.

Passive Compliance and Alignment

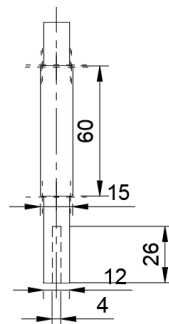
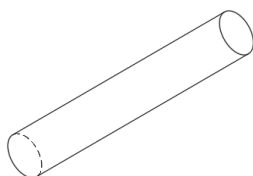
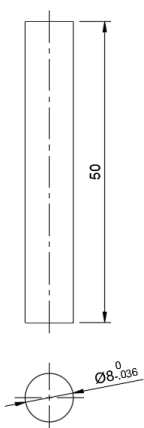
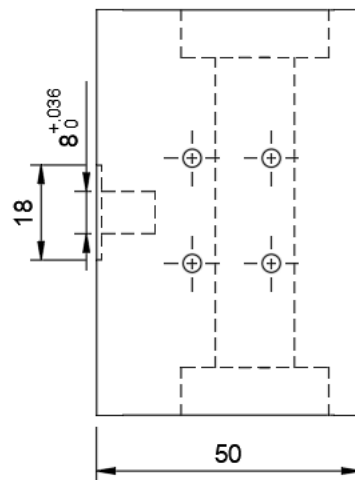
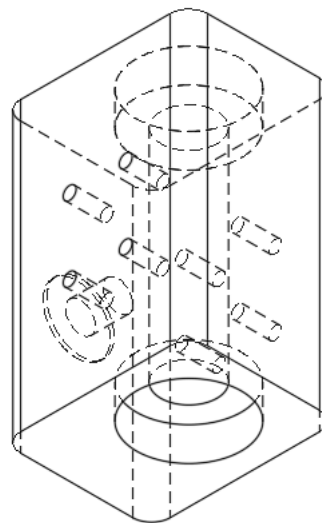
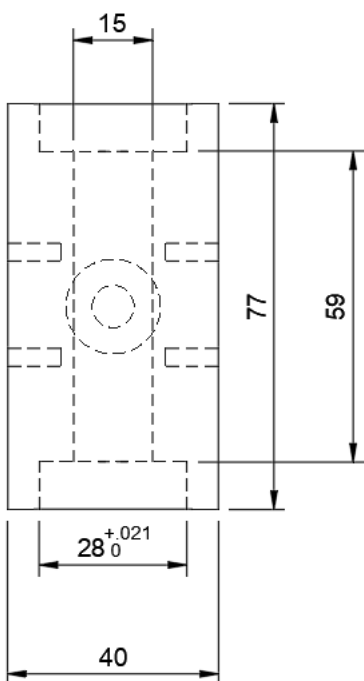
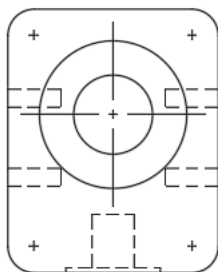
The core of the alignment is provided by the spring force:

- **Force Application:** A high-strength **compression spring** is seated between a fixed stop on the main chassis and a spring seat on the rear face of the sliding block.
- **Alignment Function:** The spring is assembled under initial compression, providing a **constant, calculated radial force** that pushes the entire sliding block and pinion assembly outward onto the rack.
- **Tolerance Compensation:** This continuous force ensures the pinion automatically tracks the rack's surface. If the rack moves slightly away, the spring expands to maintain contact; if the rack moves too close, the spring compresses, preventing tooth binding or jamming.

This assembly effectively isolates the rotational drive motion from the translational alignment motion, guaranteeing stable, optimal tooth engagement throughout the climbing operation. Below is a clear look at the CAD model and what each component represents in the pinion aligner mechanism.

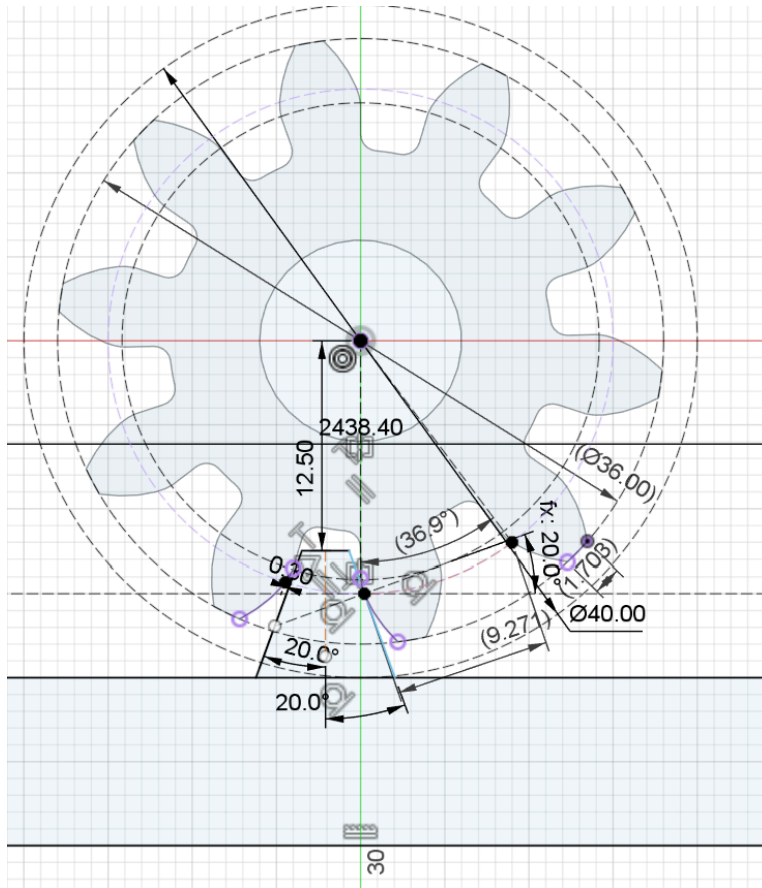


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Rack System Design & Manufacturing

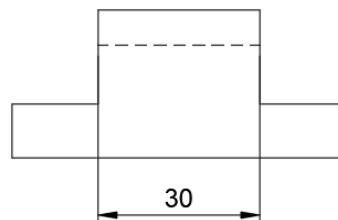
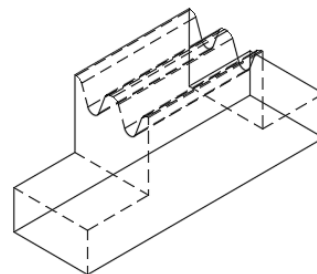
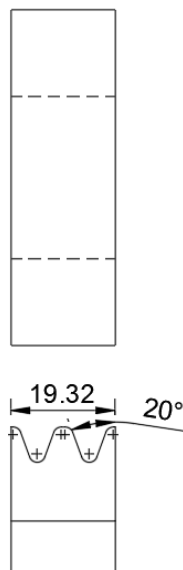
Rack Design for Manufacturing



On the left is the sketch that helped design the rack teeth. Using the 20-degree pressure angle, points of contact between pinion and rack, pitch diameter, circumference of arc between 2 contact points and more.

The rack was drawn to be 8 ft long and extruded across the gear thickness, and joints and motion links were applied for smooth animation of the pinion on the rack for both rotational and translational motion.

The rack is made of AISI 4140 steel (QT). 4140 is a chromium-molybdenum alloy steel known for its high hardenability. When subjected to **quenching and tempering** heat treatment, it achieves high hardness (wear resistance) and high core strength (load capacity).



Specification	Formula	Value (mm)	Tolerance (mm)
Module	m	3	±0.015
Circular Pitch	$\pi * m$	9.421	±0.03
Pressure Angle	Standard Spur Gear	20 degrees	±0.1 degrees
Addendum	m	3	±0.03
Dedendum	$1.25 * m$	3.75	±0.03

Rack Manufacturing Process

Cut 1.6 ft segments of the rack and the base of the rack as given in the design for attachment of 2 segments using dowel pins and bolts. This can be done by sawing or machining the segments as described in the drawings. To cut the teeth, use form milling shaped for cutting 3 mm module teeth, which cuts the entire tooth profile as the segment passes underneath. The fillet/grooves must be CNC machined, while the holes for the pins and bolts must be drilled and precision reamed. Use heat treatment to get wear resistance and hardness, and finish it all by grinding.

Quality control must be done on the teeth to check for consistent pitch, module, and pressure angle. CMM (Coordinate Measuring Machine) will probably be the best tool to check for consistent and right values of these parameters. Use the Rockwell Hardness Tester for checking the hardness of the surface and if it's within the desired range. The joint alignment holes must be precise; holes of one segment must match the holes of the other for a perfect fit and not inconsistencies in the rack profile. Flatness of the rack should also be tested; there can't be any tooth moving inwards or outwards, as the pinion might face problems, so using a laser tracker, check for straightness.

We are producing 25 linear feet of rack, which equals 15.625 segments (25 ft / 1.6 ft per segment). We'll round this up to **16 segments** for the batch.

Manufacturing Process	Time per Segment (min)	For 16 Segments (min)
Sawing/Machining	10	160
Form Milling (Teeth)	40	640
Drilling/Reaming	15	240

Heat Treatment	5	80
Grinding	10	160
Quality Control	5	80
Total Time	85	1360

A total of 1360 minutes, which comes up to 22 hours and 40 minutes for manufacturing 16 segments of 1.6 ft of rack, getting over a total of 25 feet of rack.

Manufacturing cost:

The material cost for AISI 4140 Steel bar (QT) per kg is 350 Rs (India Mart, must be lower when bulk ordered); hence, for 304.8 mm (1 ft) with a cross section of 40 mm x 70 mm, we get 853440 mm cubed volume. Density is 7.85 g/cm cubed; hence, we get a mass of 6.7 kg, giving us a total of 2344 Rs. All manufacturing process costs would come to 1000-2000 Rs per foot of the rack, giving us a total manufacturing cost of approximately 3500-5000 Rs per foot of the rack.

Modular Installation System

The modular connection system for the 1.6 ft rack segments is based on a precision lap joint design, engineered specifically to maintain absolute pitch continuity across the joint. Each segment is machined with an integrated tongue and groove feature at opposing ends, which provides coarse alignment and structural support against shear forces. The critical alignment is guaranteed by two 5 mm diameter dowel pins inserted through precision-reamed H7-tolerance holes flanking the gear teeth. These pins rigidly lock the pitch centreline before the joint is clamped. Finally, the connection is secured by four M8 bolts (Grade 8.8), torqued to a high specification, providing the necessary clamping force to prevent any micro-slippage that would compromise the rack's continuous dimensional accuracy under the robot's dynamic climbing load.

Rack segment installation relies on a three-person crew. Upon initial guidance, the precision **tongue and groove** joint is engaged to achieve alignment. Final alignment is then ensured by driving two 5 mm dowel pins into their respective H7-tolerance holes, instantaneously locking the pitch centreline across the joint. The segment's verticality is verified using a **laser plummet** or digital inclinometer, and the crucial cross-column pitch is confirmed using a **go/no-go gauge** before the four M8 bolts are torqued to spec, securing the rack segment rigidly to the chassis.

The total installation involves 25 rack segments for one vertical path (5 columns x 5 segments/column).

1. **Chassis Erection:** Erect the primary structural steel for the 5 x 5 racking system and secure the mounting brackets.
2. **Level 1 Base Row:** Install C1R1, C2R1, C3R1, C4R1, and C5R1 (all 5 base segments).
3. **Complete Column 1:** Install C1R2, C1R3, C1R4, and C1R5 (segments 2 through 5 in Column 1). Check the verticality of C1 with the laser plummet after every segment.
4. **Complete Remaining Columns (C2-C5):** Proceed to install Column 2 (C2R2-C2R5). Before installing C2R2, check the pitch continuity between C1 and C2. Repeat for Columns C3, C4, and C5.
5. **Final QC Check:** Perform a final, end-to-end alignment and pitch check on all 5 columns to ensure the entire track is ready for robot usage.

Preparation, lifting and initial guidance will take 10 minutes per segment. Pinning the segment and checking vertical alignment will take another 10 minutes per segment. Checking if the whole column has a consistent pitch and flatness will take about 10 minutes for each column, and finally bolting it thoroughly for each segment will take about 5 minutes. Which will give us a total of $10 * 25 + 10 * 25 + 10 * 5 + 10 * 25 = 800$ minutes in total.

BOM (Per Pinion Assembly/Per Column Rack)

Component	Material	Units	Unit Cost (Rs)	Total Cost (RS)	Supplier
Sliding Block	Aluminium 6061	1	1500	1500	Machining Shop
12mm ID, 28mm OD Bearing	6001 2RS	2	150	300	Bearing Supplier (India Mart)
Linear Mounted Bearing	Steel	2	2500	5000	Bearing Supplier (India Mart)
Spring	ASTM A228 Music Wire	1	300	300	Spring Manufacturer
Pinion	AISI 8640 Steel (QT)	1	1000	1000	Gear Fabricator

Axle Shaft	AISI 4140 Steel (QT)	1	500	500	Machining Shop
Spring Guide	Stainless Steel	1	100	100	Machining Shop
Guide Rail	Stainless Steel	2	100	200	Machining Shop
Set Screw	Stainless Steel	4	20	80	Fastener Supplier
Dowel Pin	Steel	12	80	960	Fastener Supplier
M8 Bolts	Steel	24	40	960	Fastener Supplier
Key	AISI 4140 Steel (QT)	1	20	20	Machining Shop
Total Cost				10920	

Rack Assembly DFM

The rack assembly demonstrates high **Design for Assembly (DFA)** efficiency due to its modular design utilising pre-assembled 1.6 ft segments, reducing on-site welding or custom cutting. Manufacturing efficiency is maximised by using specialised **rack milling** for tooth cutting, which is scalable for high volumes, and by limiting post-heat treatment grinding only to the critical joint faces. However, the design faces two primary assembly challenges: first, maintaining **pitch continuity** across the lap joint is highly dependent on the precision of the 5 mm dowel pin holes, requiring rigorous quality control and skilled installers. Second, achieving and verifying the correct **cross-column pitch** on-site, especially for the 5 x 5 configuration, demands the use of high-cost laser alignment tools and specialised **go/no-go gauges** to ensure the four robot pinions track correctly between columns, which adds complexity and time to the installation phase.